

# AWS IAM Access Analyzer

## To Begin with the Lab

### Summary of the Lab

In this lab, **AWS IAM Access Analyzer** was introduced as a security reporting feature used to review resource permissions inside an AWS account. The video explained that IAM Access Analyzer is used to identify resources that are shared with external entities and permissions that are granted but never used. It also highlighted that the analyzer is created per AWS Region, unlike some IAM reports that are global. The lab showed how to create an analyzer and review external access findings, such as a cross-account role or publicly shared resource. Finally, the lesson explained that unused access findings help reduce unnecessary permissions and improve the overall security posture.

### Understanding AWS IAM Access Analyzer

**AWS IAM Access Analyzer** is a security analysis feature that helps you understand who can access your AWS resources and whether some permissions are not being used. It exists so administrators can detect risky sharing, such as public access or cross-account access, before it becomes a problem. It also helps identify unused permissions so you can tighten security and follow the principle of least privilege. Unlike the credential report, Access Analyzer is shown inside the **IAM** console and can be created for a specific AWS Region.

#### Example:

An **Amazon S3** bucket is accidentally shared with an external AWS account through a cross-account role. **AWS IAM Access Analyzer** flags that role as an external access finding. Later, if another role has permissions that have never been used, Access Analyzer can also flag that as unused access so it can be reviewed or removed.

### Lab Steps

#### Step 1 – Open IAM Access Analyzer

- Sign in to the AWS Management Console.
- Open **IAM**.
- Navigate to the **Access analyzer** section.

**Identity and Access Management (IAM)**

Dashboard

▼ **Access management**

- User Groups
- Users
- Roles
- Policies
- Identity providers
- Account settings
- Root access management **New**
- Temporary delegation requests

▼ **Access reports**

- Access analyzer**
- Resource analysis **New**
- Unused access
- Analyzer settings
- Credential report
- Organization activity
- Service control policies
- Resource control policies

**Security, Identity & Compliance**

## IAM Access Analyzer

Simplify your journey to least privilege

**How it works**

- 1 Create an analyzer**  
Create a powerful analyzer to monitor external, internal, and unused access patterns across your AWS environment, whether for an entire organization or individual accounts.
- 2 Review findings**  
IAM Access Analyzer automatically identifies access patterns across your AWS environment, detecting both external access from outside your zone of trust and internal access. IAM Access Analyzer generates comprehensive findings about unused access for IAM users and roles, helping you maintain least-privilege access and strengthen your security posture.
- 3 Take action**  
If the access is intended, you can archive the finding so that you can focus on reviewing active findings. If the access is not intended, you can resolve the finding by removing the unused access or by modifying the policy to remove access to the resource.

**Get started**

- External access findings**  
Identify resources shared with an external entity
- Internal access findings**  
Identify internal IAM users and roles that can access specific resources
- Unused access findings**  
Identify IAM users and roles with unused access

**Create analyzer**

**Pricing (USD)**

|                                   |
|-----------------------------------|
| <b>External access</b>            |
| Free                              |
| <b>Internal access</b>            |
| \$9 / AWS resource / month        |
| <b>Unused access</b>              |
| \$0.2 / IAM user and role / month |

[AWS Pricing Calculator](#)

## Step 2 – Create an External Access Analyzer

- Click **Create analyzer**.

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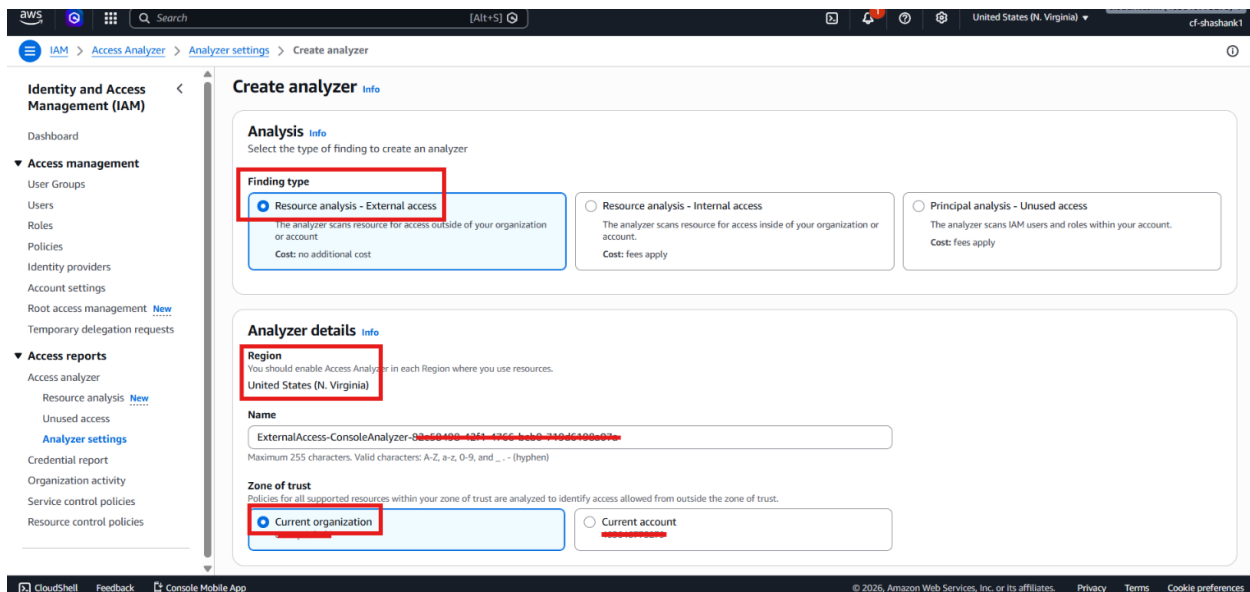
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[AWS Pricing Calculator](#)

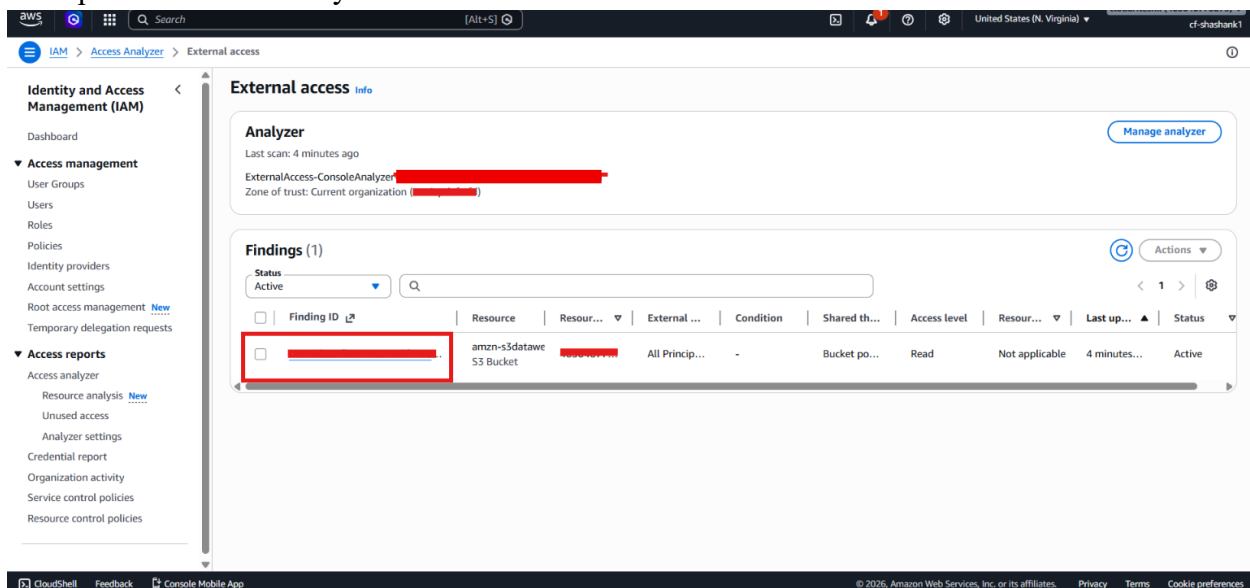
- Select **External access analysis**.
- Choose the AWS Region for the analyzer.
- For Zone of Trust, I am choosing my own organization.
- Click **Create analyzer**.



- Wait for the analyzer to become active as it will take around 1-2 mins depending on usage and permissions the account have.

## Step 3 – Review External Access Findings

- Open the created analyzer.



- Review the **External access findings** tab.
- Check for resources shared outside your AWS account.

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**Finding details**

Public: this finding is for a resource that allows public access.

**Details**

|                               |                              |                                                  |                     |
|-------------------------------|------------------------------|--------------------------------------------------|---------------------|
| <b>Finding ID</b>             | <b>External principal (I</b> | <b>Resource control policy (RCP) restriction</b> | <b>Access level</b> |
|                               | All Principals               | Not applicable                                   | Read                |
| <b>Resource</b>               | <b>Condition</b>             | <b>Updated</b>                                   | • s3:GetObject      |
|                               | -                            | 7 minutes ago                                    |                     |
| <b>Resource owner account</b> | <b>Bucket policy</b>         | <b>Status</b>                                    |                     |
|                               |                              | Active                                           |                     |

**Next steps**

**Intended access**

If the access is intended, such as access necessary for business processes, you can archive the finding. This lets you focus on findings that are related to potential security risks. When you archive a finding, it's removed from Active findings and the status changes to Resolved.

[Archive](#)

To automatically archive similar findings, [create an archive rule](#).

**Not intended**

If the access isn't intended, it indicates a potential security risk. Use the console for the service associated with the resource to modify or remove the policy that grants the unintended access. To confirm that your change removed the access, choose **Rescan**. If the access is removed, the status changes to Resolved.

[Go to S3 console](#)

arn:aws:s3::arn:s3-datowebyear1

- Identify cross-account roles, public resources, or other external principals.
- Review the ARN of the resource that is exposed.
- I don't have any cross-account finding for now so it is empty but it will show up here.

## Step 4 – Review Unused Access Findings

- Open the **Unused access findings** tab.
- Review permissions that have not been used.
- Identify IAM users, groups, or roles with unused access.
- Use the findings to decide which permissions should be removed.
- Note that creating unused access analysis has a cost that's why I have not created it here for now.

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**Unused access**

**Analyzer**

Unused access analyzer

No analyzer is set up. [Learn more](#)

[Create analyzer](#)

## **Step 5 – Use the Findings for Security Review**

- Confirm whether external access is intended or accidental.
- Remove unnecessary sharing if required.
- Remove unused permissions when possible.
- Use the report to improve security posture and follow least privilege.

## **Verification**

- **AWS IAM Access Analyzer** was reviewed successfully.
- External access findings were identified from the **IAM** console.
- Unused permissions could be reviewed for cleanup.
- The analyzer worked at the Region level.
- The report helped confirm shared resources and unused access risks.